

Battery Design & Test Guidance

SOW dated 25 Nov 2024

- 1.0 Objective – To guide the final design, perform design analysis, to assess margins for thermal runaway propagation resistance of batteries. This includes the same design guidance for our medium size calorimeter.
- 2.0 Scope – The scope includes design development from a concept design to a preliminary design review maturity of 3 passively propagation resistant (PPR) battery design and maturation of a medium sized thermal runaway calorimeter. Two of the designs will use the 21700 cell format and the third will use 46mm diameter cell designs.
- 3.0 NASA Inputs – NASA to provide the CAD models for the battery designs. NASA will provide thermal runaway calorimetry results.
- 4.0 Contractor Tasks
 - 4.1.1 .
 - 4.2 Scale up 18650 Al Spine Heat Sink Battery Design to 21700
 - 4.2.1 Test Results Review and Model Anchoring – Using NASA PPR test results as inputs, make adjustments to the models used in 4.2.1 shall be made to enable better predictions of margins for protecting adjacent cells.
 - 4.3 Highest Specific Energy PPR Battery Design (M4)
 - 4.3.1 Design & Analysis Trades – This design will utilize the Rincell RC41 18650 cell design achieving 300 Wh/kg and seek to maximize specific energy. Lighter composite heat sink materials, new cell to heat sink interface materials, and new features to protect adjacent cells from ejecta will need thermal analysis for providing design guidance. The analysis shall focus on predicting margins for protecting adjacent cells.
 - 4.3.2 Test Results Review and Model Anchoring – Using NASA PPR test results as inputs, make adjustments to the models using in 4.6.1 shall be made to enable better predictions of margins for protecting adjacent cells.
 - 4.3.3 Reporting – Present the trades, analyses, and model enhancements at the Power Sources Conference in 2025.
 - 4.4 Medium Fractional Thermal Runaway Calorimeter
 - 4.4.1 Design & Analysis Trades – Delivering heat to these big cells (46mm dia x 95mm long) to trigger them into thermal runaway in under 3 minutes to enable good signal to noise calorimetry requires trading heaters, insulative materials, and cell chamber designs.
 - 4.4.2 Test Results Review and Model Anchoring – Use NASA test results to anchor the model predictions in 4.7.1 to guide design towards quicker triggering and better calorimetry.
 - 4.5 PPR Battery Design with 46mm Cell Designs
 - 4.5.1 Design & Analysis Trades – Using the lessons learned from M3, M4, M5, and M6, guide the design through cell spacing, heat sinking, electrical busing, and ejecta management trades to achieve PPR while also allowing C/3 charge and C/3 discharge performance without excessive cell to cell thermal gradients.

- 4.5.2 Test Results and Model Anchoring – Use NASA PPR test results to make adjustments to the models in 4.8.1 to enable better understanding and margin predictions for the adjacent cells.

5.0 Deliverables and Period of Performance

5.1 Spine Heat Sink Battery Design

- 5.1.1 Test results review and model anchoring (1-months ARO)

5.2 Highest Specific Energy PPR Battery Design

- 5.2.1 Design and assembly trades for subscale (1-month ARO)
- 5.2.2 Test results review and model anchoring (6-months ARO)
- 5.2.3 Reporting (8-months ARO)

5.3 Medium Fractional Thermal Runaway Calorimeter

- 5.3.1 Design and analysis trades (3-months ARO)
- 5.3.2 Test results review and model anchoring (6-months ARO)

5.4 PPR Battery Design with 46mm Cell Designs

- 5.4.1 Design and analysis trades (9-months ARO)
- 5.4.2 Test results and model anchoring (12-months ARO)